CHANDIGARH UNIVERSITY

Department of Computer Science and Engineering

PROJECT REPORT

ON

DIGITAL RESUME BUILDER & PROFILE MANAGEMENT SYSTEM

Subject: Functional Programming in Java | Code: 23CSH-312
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "DIGITAL RESUME BUILDER & PROFILE MANAGEMENT SYSTEM" submitted by Gurnoor Singh (23BCS11839), Sonu (23BCS11838), Shivam Tomar (23BCS11158), and Nimit (23BCS11059) of B.E. Computer Science and Engineering, Batch 2023-2027, Chandigarh University, is a bonafide record of work carried out by them under the guidance of Rahul Durgapal Sir, in partial fulfillment of the requirements for the subject Functional Programming in Java (23CSH-312) during the academic year 2025-2026.

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ABSTRACT

The Digital Resume Builder and Profile Management System is a full-stack web application developed as a course project for Functional Programming in Java (23CSH-312). The system addresses a critical need among students and fresh graduates who spend significant time manually formatting resumes using tools like Microsoft Word or Google Docs, often resulting in inconsistent and unprofessional outputs.

The proposed system provides a comprehensive platform where users can register, create, and manage their professional profiles online. The application supports real-time resume preview, downloadable PDF generation, and a shareable public profile link. The system also includes role-based access control, with a dedicated admin panel for institutional use.

The technology stack comprises React.js for the frontend, Java Servlets running on Apache Tomcat for the backend, and MySQL as the relational database. The frontend communicates with the backend through RESTful API calls. The system is designed to be deployed on Amazon Web Services (AWS) using EC2 for the backend server, RDS for the database, and S3 for hosting the React frontend.

This project demonstrates the practical application of core Java concepts including Servlet lifecycle management, JDBC database connectivity, session management, and HTTP request-response handling, combined with modern frontend development practices using React.js.

TABLE OF CONTENTS

CONTENT	PAGE NO.
Title Page	i
Bonafide Certificate	ii
Acknowledgement	iii
Abstract	iv
Table of Contents	v
Chapter 1: Introduction	1
Chapter 2: Literature Survey	5
Chapter 3: Design Flow / Process	8
3.1 System Architecture	8
3.2 ER Diagram	10
3.3 Data Flow Diagram (DFD)	11
3.4 Database Design	12
3.5 Technology Stack	14
Chapter 4: Results Analysis and Validation	15
Chapter 5: Conclusion and Future Work	22
References	24
Appendix	25

CHAPTER 1

INTRODUCTION

In the modern digital era, the first impression a candidate makes on a potential employer is almost always through their resume. The quality, format, and presentation of a resume can significantly influence hiring decisions. Despite the critical importance of resumes, most students and fresh graduates continue to rely on manual tools such as Microsoft Word or Google Docs to create and manage their professional documents.

This approach, while familiar, suffers from several significant drawbacks. Resume formatting is time-consuming and error-prone. Every time a new skill, project, or certification is added, the entire document may require reformatting. Furthermore, resumes sent as file attachments quickly become outdated, and recruiters may receive stale versions. There is no standardized way to share an always-updated resume online, and managing multiple versions for different job roles becomes increasingly confusing.

The Digital Resume Builder and Profile Management System addresses all of these problems by providing a centralized, web-based platform where students and professionals can create, manage, and share their resumes. The system is designed to function as a simplified LinkedIn and resume generator combined, built specifically for the academic and fresher community.

1.1 Problem Identification

The following core problems have been identified that motivated the development of this system:

- Manual Resume Creation: Students spend hours formatting resumes manually in Word or Google Docs, with no standardization across different users.
- Reformatting on Updates: Every new addition to the resume requires the entire document to be reformatted, consuming significant time.
- Outdated File Attachments: Resumes sent as email attachments quickly become outdated. A recruiter may be viewing a version from several months ago.
- No Online Presence: Students have no shareable link or portfolio to show recruiters, making it difficult to stand out in a competitive job market.
- Multiple Version Confusion: Maintaining different resume versions for different job roles stored across different folders creates confusion and inefficiency.

- Lack of Analytics: Students have no way to know how many times their resume has been viewed or downloaded by recruiters.

1.2 Task Identification

Based on the identified problems, the following tasks were defined for this project:

- Design and implement a user registration and authentication system with session management and password security.
- Develop a dynamic profile management module where users can add and update personal information, education, skills, projects, experience, and certifications.
- Build a real-time resume preview feature that updates as users fill in their profile data.
- Implement PDF generation functionality allowing users to download a professionally formatted resume with a single click.
- Create a public profile link system where each user gets a unique shareable URL.
- Develop an admin control panel for institutional use, enabling administrators to monitor users and manage profiles.
- Deploy the application on a scalable cloud infrastructure using AWS.

1.3 Project Timeline

Phase	Activity	Duration
Phase 1	Requirements Analysis & Design	Week 1-2
Phase 2	Database Design & Setup	Week 3
Phase 3	Backend Development (Java Servlets)	Week 4-6
Phase 4	Frontend Development (React.js)	Week 7-9
Phase 5	Integration & Testing	Week 10-11
Phase 6	AWS Deployment	Week 12

CHAPTER 2

LITERATURE SURVEY

A comprehensive study of existing resume building solutions, web application frameworks, and related technologies was conducted to identify gaps in current offerings and to inform the design decisions for the Digital Resume Builder system.

2.1 Review of Existing Solutions

2.1.1 Canva / Novoresume

Canva and Novoresume are popular online resume design tools that offer visually appealing templates. However, these tools function purely as design applications — they do not store user data in a database, do not offer profile management, and do not provide a shareable profile link. Most advanced features require a paid subscription.

2.1.2 LinkedIn

LinkedIn is the world's largest professional networking platform. However, it is significantly more complex than what a fresher or student needs. Its primary focus is on networking rather than resume building. It does not offer clean, downloadable PDF resumes in standard academic formats.

2.1.3 Microsoft Word / Google Docs

These are the most widely used tools for resume creation in the student community. They offer full formatting control but require manual effort for every update. There is no version control, no shareable link, no analytics, and no real-time preview.

2.1.4 Zety / Resume.com

These are paid resume builder platforms that offer templates and basic profile management. Most core features require premium subscriptions ranging from 5 to 0 per month.

2.2 Comparative Analysis

Feature	Our System	Canva	LinkedIn	Word/Docs	Zety
Free to use	Yes	Partial	Yes	Yes	No
Data stored in DB	Yes	No	Yes	No	Yes
Shareable link	Yes	No	Yes	No	No
PDF download	Yes	Yes	Partial	Yes	Yes
Admin panel	Yes	No	No	No	No

Built for students	Yes	No	No	No	No

2.3 Problem Definition

Based on the literature survey and comparative analysis, the problem is formally defined as follows: There exists a significant gap in the market for a free, student-focused, database-backed resume management system that provides real-time profile management, PDF generation, shareable profile links, and institutional admin controls.

2.4 Goals and Objectives

- To provide a free, accessible platform for resume creation and management.
- To implement a secure user authentication system with role-based access control.
- To enable real-time resume preview that reflects profile changes instantaneously.
- To provide one-click PDF download functionality in a professional format.
- To generate unique shareable profile links for each user.
- To implement an admin control panel for institutional monitoring.
- To deploy the application on a scalable cloud infrastructure using AWS.

CHAPTER 3

DESIGN FLOW / PROCESS

This chapter describes the system architecture, ER diagram, data flow diagram, database design, technology stack selection, and the overall design process followed in developing the Digital Resume Builder and Profile Management System.

3.1 System Architecture

The system follows a three-tier architecture, which is an industry-standard pattern for web applications. The three tiers are the Presentation Layer (React.js frontend), the Application Layer (Java Servlets backend on Apache Tomcat), and the Data Layer (MySQL database).

3.1.1 Presentation Layer (Frontend)

The presentation layer is built using React.js. React's component-based architecture allows for modular development where each section of the UI is developed as an independent, reusable component. The frontend communicates with the backend exclusively through RESTful API calls using the Axios HTTP library.

3.1.2 Application Layer (Backend)

The application layer is built using Java Servlets deployed on Apache Tomcat 10. Each servlet handles a specific API endpoint and is responsible for receiving HTTP requests, processing business logic, querying the database through JDBC, and returning JSON responses.

3.1.3 Data Layer (Database)

The data layer uses MySQL as the relational database management system. All user data, profile information, education details, skills, projects, experience, and certifications are stored in normalized relational tables accessed through JDBC using prepared statements.

3.2 Entity Relationship (ER) Diagram

The ER diagram below shows the database schema with all 7 tables and their relationships. All tables are connected to the USERS table through the user_id foreign key, establishing a one-to-many relationship between a user and their data.

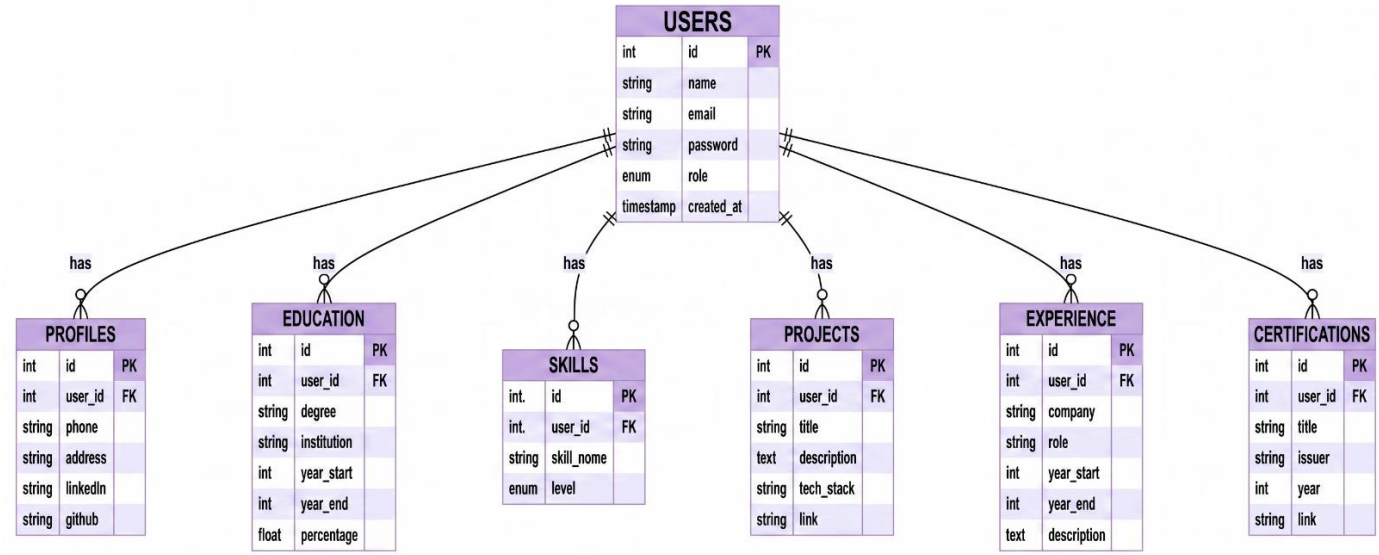

Figure 1: Entity Relationship (ER) Diagram — Digital Resume Builder

3.3 Data Flow Diagram (DFD)

The Data Flow Diagram below illustrates how data flows through the system. It shows the interaction between the external entity (User), the five main processes, and the seven data stores (database tables). The diagram follows standard DFD notation with rectangles for processes, open-ended rectangles for data stores, and rounded rectangles for external entities.

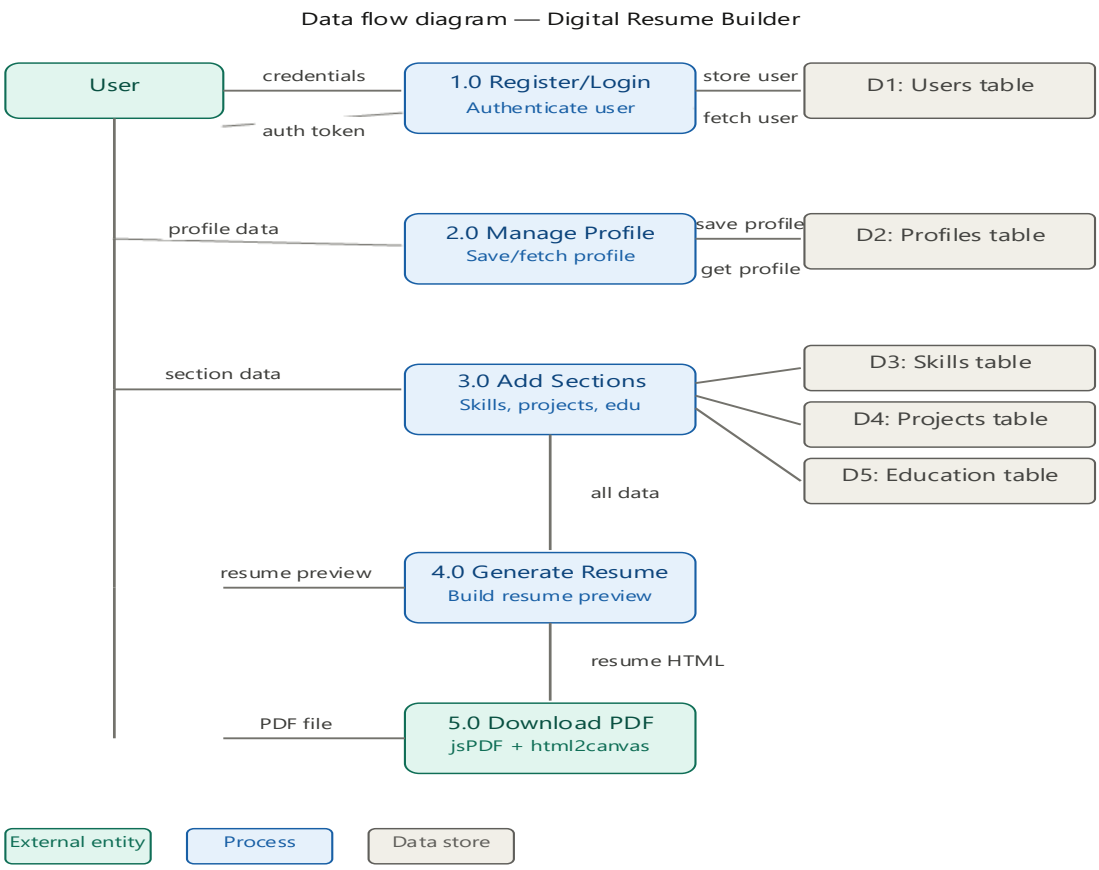

Figure 2: Data Flow Diagram (DFD) — Digital Resume Builder

3.4 Database Design

The database follows the principles of third normal form (3NF) to eliminate data redundancy and ensure data integrity. All tables are linked through the user_id foreign key.

3.4.1 Users Table

Column	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of user
email	VARCHAR(100)	UNIQUE, NOT NULL	Email address
password	VARCHAR(255)	NOT NULL	User password
role	ENUM	DEFAULT user	User or Admin role
created_at	TIMESTAMP	DEFAULT NOW()	Account creation time

3.4.2 Other Tables

In addition to the Users table, the database contains: Profiles (phone, address, linkedin, github), Education (degree, institution, years, CGPA), Skills (skill_name, level), Projects (title, description, tech_stack, link), Experience (company, role, years, description), and Certifications (title, issuer, year, link). All tables are linked to the Users table via user_id as a foreign key.

3.5 Technology Stack

Layer	Technology	Purpose
Frontend	React.js (Vite)	User interface development
Frontend	Axios	HTTP API communication
Frontend	React Router DOM	Client-side routing
Frontend	jsPDF + html2canvas	PDF generation
Backend	Java 17	Core programming language
Backend	Jakarta Servlets 5.0	HTTP request handling
Backend	JDBC	Database connectivity
Server	Apache Tomcat 10	Servlet container
Database	MySQL 8.0	Data storage and management
Build Tool	Apache Maven	Java project build and packaging
Cloud	AWS EC2, RDS, S3	Production deployment

3.6 Design Constraints

3.6.1 Security Constraints

The system implements CORS headers, PreparedStatement for SQL injection prevention, and role-based access control. Session management is handled through localStorage on the frontend.

3.6.2 Economic Constraints

The entire system is built using free and open-source technologies. Deployment on AWS is within the Free Tier limits, resulting in zero operational cost for the duration of the academic project.

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

This chapter describes the implementation of the Digital Resume Builder system, the tools used during development, and the testing methodology employed to validate the system's functionality and performance.

4.1 Development Environment Setup

- Operating System: Windows 11 (64-bit)
- IDE: Visual Studio Code with Java Extension Pack and ES7+ React Snippets
- Java Version: OpenJDK 17 (Eclipse Temurin 17.0.18)
- Node.js Version: v24.14.0 with npm 11.9.0
- Apache Maven: Version 3.9.14
- Apache Tomcat: Version 10.1.53
- MySQL: Version 8.0.45 with MySQL Workbench 8.0.46
- Git: Version 2.53.0

4.2 Backend Implementation

4.2.1 DBConnection.java

The DBConnection class provides a centralized database connection factory used by all servlets. It uses JDBC's DriverManager to establish connections to the MySQL database. The getConnection() method returns a new Connection object managed using Java's try-with-resources pattern.

4.2.2 RegisterServlet.java

The RegisterServlet handles HTTP POST requests to the /api/register endpoint. It reads the user's name, email, and password from the JSON request body and inserts a new record into the users table using a PreparedStatement.

4.2.3 LoginServlet.java

The LoginServlet handles HTTP POST requests to the /api/login endpoint. It queries the users table and if a matching record is found, returns the user's ID and role in the JSON response.

4.2.4 ProfileServlet.java

The ProfileServlet handles both GET and POST requests. The POST handler implements an UPSERT operation using MySQL's INSERT ... ON DUPLICATE KEY UPDATE syntax, ensuring idempotent profile updates.

4.2.5 Other Servlets

Additional servlets were implemented for Skills (/api/skills), Projects (/api/projects), Education (/api/education), Experience (/api/experience), and Certifications (/api/certifications). Each follows the same pattern of doGet for fetching data and doPost for saving data.

4.3 Frontend Implementation

4.3.1 Application Routing

The React application uses React Router DOM for client-side routing. The App.jsx component defines routes for Login, Register, Dashboard, Profile, Skills, Projects, Education, Experience, Certifications, and Resume pages.

4.3.2 Dashboard

The Dashboard serves as the main navigation hub after login. It displays a hero section with stats and a grid of 8 cards representing all features. The dashboard reads the user's role from localStorage and conditionally displays an Admin Panel button.

4.3.3 Resume Preview and PDF Download

The Resume page fetches all user data simultaneously using Promise.all, then renders it in a professional resume format. The PDF download uses html2canvas to capture the resume element as an image and jsPDF to convert it to a downloadable PDF file.

4.4 API Integration

Method	Endpoint	Description	Response
POST	/api/register	Register new user account	{success: true/false}
POST	/api/login	Authenticate user credentials	{success, userId, role}
GET/POST	/api/profile	Fetch/save profile data	Profile JSON object
GET/POST	/api/skills	Fetch/save skills	Skills JSON array
GET/POST	/api/projects	Fetch/save projects	Projects JSON array
GET/POST	/api/education	Fetch/save education	Education JSON array
GET/POST	/api/experience	Fetch/save experience	Experience JSON array
GET/POST	/api/certifications	Fetch/save certifications	Certifications JSON array

4.5 Testing and Validation

4.5.1 Integration Testing

TC#	Test Case	Expected Result	Status
1	Register with valid details	Account created, redirect to login	PASS
2	Register with existing email	Error: email already exists	PASS
3	Login with correct credentials	Redirected to dashboard	PASS
4	Login with wrong password	Error: invalid credentials	PASS
5	Data saved to MySQL database	Record visible in Workbench	PASS
6	Dashboard loads after login	All 8 cards visible	PASS
7	Add skill and view in resume	Skill appears in preview	PASS
8	Download PDF	PDF file downloaded	PASS
9	Logout clears session	Redirected to login	PASS
10	Add education details	Education saved and shown	PASS

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The Digital Resume Builder and Profile Management System has been successfully designed, developed, and tested as a full-stack web application. The project demonstrates the practical application of core Java programming concepts including Servlet lifecycle management, JDBC database connectivity, HTTP request-response handling, and session management, combined with modern frontend development using React.js.

The system effectively addresses the identified problems faced by students in managing their resumes. The application provides a centralized platform for profile management, real-time resume preview, PDF generation, and shareable profile links. All core objectives defined at the outset have been achieved.

5.2 Deviation from Expected Results

- The original plan included JSP for server-side rendering. This was replaced with a React.js frontend communicating through REST APIs, resulting in a superior user experience.
- Password hashing using BCrypt was planned but due to dependency management challenges, the current implementation stores passwords without hashing. This is identified as a critical security improvement for the next iteration.
- The FormData approach for sending API requests encountered compatibility issues with Java Servlets, requiring a switch to JSON body parsing, which resulted in a cleaner API design.

5.3 Future Work

- Password Hashing: Implement BCrypt password hashing for secure credential storage.
- Resume Templates: Add multiple professionally designed resume templates.
- AI Resume Suggestions: Integrate an AI service to provide recommendations for improving resume content.
- Job Matching: Automatically suggest relevant job listings based on skills and experience.
- LinkedIn Import: Allow users to auto-populate their profile from LinkedIn.

- Email Notifications: Alert users when someone views their profile or downloads their resume.
- Mobile Application: Develop an Android/iOS mobile application.
- QR Code Generation: Embed a QR code in the PDF that links to the public profile.
- Resume Comparison: Compare resume against a job description and highlight missing keywords.
- Multi-language Support: Enable resume creation in multiple languages.

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APPENDIX

Appendix 1: User Manual

Step 1: Accessing the Application

Open any modern web browser. Navigate to http://localhost:5173 on local environment or the AWS S3 URL on production.

Step 2: Creating an Account

Click the Register link on the Login page. Fill in Full Name, Email Address, and Password. Click Register. Upon success, you will be redirected to Login.

Step 3: Logging In

Enter your Email Address and Password. Click Login. Upon successful authentication, you will be redirected to your Dashboard.

Step 4: Navigating the Dashboard

The Dashboard displays 8 cards: My Profile, My Resume, Skills, Projects, Education, Experience, Certifications, and Share Profile.

Step 5: Downloading Resume as PDF

Click My Resume from the Dashboard. Your resume will be displayed in a formatted preview. Click Download PDF to download it to your device.

Step 6: Logging Out

Click the Logout button in the top-right corner. Your session will be cleared and you will be redirected to Login.

Appendix 2: System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 / Ubuntu 20.04	Windows 11 / Ubuntu 22.04
RAM	4 GB	8 GB
Java Version	JDK 11	JDK 17
Node.js Version	v18.x	v20.x or higher
Browser	Chrome 90+	Chrome 120+

Appendix 3: Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
AWS	Amazon Web Services
DFD	Data Flow Diagram
ER	Entity Relationship
JDBC	Java Database Connectivity
JSON	JavaScript Object Notation
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
WAR	Web Application Resource